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Gestures Recognition

MLSM2154 : Artificial Intelligence course

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Résumé

Hand gesture recognition is a central problem in Human–Computer Interaction (HCI), where the goal is to classify sequences of three-dimensional coordinates $\mathbf{r}(t) = [x(t), y(t), z(t)]^\top$ recorded over discrete time. This project implements and compares four classifiers on the gesture datasets introduced by Huang et al. [5], namely *domain 1* (digits 0–9) and *domain 4* (3D figures). Two baseline methods based on dynamic programming are first considered : Dynamic Time Warping (DTW) and an Edit-distance-based classifier, the latter relying on a preliminary vector-quantization step. Two state-of-the-art approaches are then investigated : the Three-Cents (3) recognizer, a 3D adaptation of the \$1 recognizer by Wobbrock et al. [13], and a Recurrent Neural Network (RNN). All methods are evaluated through leave-one-user-out (user-independent) and leave-one-gesture-sample-out (user-dependent) cross-validation procedures, and compared using hypothesis testing. This report details the theoretical framework of each method, the implementation strategy, and a rigorous statistical comparison of their accuracies.

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1 Introduction

The development of natural user interfaces has fostered a growing interest in hand gesture recognition as a means of interaction between humans and machines. Gestures, being rapidly executed freehand drawings [5], exhibit a strong inherent variability : two instances of the same gesture produced by the same user may differ in speed, scale, spatial position, and duration. The sequential nature of the signal and this variability are the main challenges any recognizer must address.

In this work we consider the problem of classifying three-dimensional gesture trajectories recorded as a sequence of spatial coordinates. Following the project guidelines, we assume constant time steps ($t = 1, 2, 3, \dots$) and ignore the precise sampling instants. Four classifiers are implemented and compared :

1. Dynamic Time Warping (DTW) as a first baseline, combined with a k-nearest-neighbors (k-NN) classifier ;
2. Edit-distance on quantized symbolic sequences, as a second baseline ;
3. The Three-Cents (3) recognizer, a 3D extension of the \$1 recognizer [13] ;
4. A Recurrent Neural Network (RNN) trained end-to-end.

Each method is evaluated on two datasets from [5] under both a *user-independent* and a *user-dependent* cross-validation protocol. The research questions underlying this project are threefold : (Q1) *how do classical template-matching baselines compare with more recent recognizers on 3D mid-air gestures ?* ; (Q2) *do user-independent settings systematically degrade accuracy, as expected ?* ; and (Q3) *is one of the investigated classifiers significantly better than the others in the user-independent setting ?*

The remainder of the report is organized as follows. Section 2 describes the datasets. Section 3 develops the theoretical framework of the four methods. Section 4 outlines the implementation and evaluation methodology. Section 5 presents and discusses the experimental results, and Section 6 concludes.

2 Data description

All the data used in this project come from the experiment of Huang et al. [5]. The complete dataset comprises four domains of ten gestures each, each gesture being performed ten times by ten different subjects, for a total of $10 \times 10 \times 10 = 1000$ samples per domain. In this work we focus on two of these domains :

— Domain 1 : hand-drawn representations of the digits $\{0, 1, \dots, 9\}$ in 3D ;

— Domain 4 : 3D geometric figures.

Each gesture is recorded as a finite sequence of 3D position vectors

$$\mathbf{r}(t) = [x(t), y(t), z(t)]^\top \in \mathbb{R}^3, \quad t = 1, 2, \dots, T_i, \quad (1)$$

where T_i is the length of the i-th sample and may vary across recordings. As stated in the project guidelines, the discrete time index is treated as uniform, even though this assumption is not strictly satisfied in practice.

3 Theoretical framework

This section introduces the four methods used in our comparative study. We first describe the two dynamic-programming baselines (DTW and Edit-distance), then the Three-Cents recognizer, and finally the Recurrent Neural Network. Throughout, we denote the test sequence by $\mathcal{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$ and a reference (template) sequence by $\mathcal{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_M)$, with $\mathbf{x}_i, \mathbf{y}_j \in \mathbb{R}^3$.

3.1 Dynamic Time Warping (DTW)

Dynamic Time Warping is a classical technique for comparing two time-dependent sequences that may differ in speed or duration [9]. The key idea is to find a non-linear alignment between the two sequences that minimizes a cumulative distance, thereby absorbing local temporal distortions.

We first define the local cost between two points $\mathbf{x}_i$ and $\mathbf{y}_j$ via the Euclidean (L_2) distance :

$$d(i, j) = \|\mathbf{x}_i - \mathbf{y}_j\|_2. \quad (2)$$

The accumulated distance $D(i, j)$ is then computed recursively using dynamic programming [10] :

$$D(i, j) = d(i, j) + \min \begin{cases} D(i-1, j-1) \\ D(i-1, j) \\ D(i, j-1) \end{cases} \quad (3)$$

with boundary condition $D(1, 1) = d(1, 1)$. The three candidate predecessors correspond respectively to a diagonal step (temporal alignment), a vertical step (the test sequence is progressing while the reference is stalling), and a horizontal step (the reference is progressing while the test is stalling).

A warping path $W = (w_1, \dots, w_L)$, with $w_k = (i_k, j_k)$, is a sequence of grid cells traversed from $(1, 1)$ to (N, M). To be physically meaningful, W must satisfy three standard constraints [10] :

— Boundary : $w_1 = (1, 1)$ and $w_L = (N, M)$;

— Monotonicity : both index sequences (i_k) and (j_k) are non-decreasing (temporal order is preserved) ;

— Continuity : transitions occur only between adjacent cells, so that no observation is skipped.

The final similarity score $D(N, M)$ is then plugged into a k-nearest-neighbors classifier : the test gesture is assigned the majority label among its k closest templates.

3.2 Edit-distance on quantized sequences

The second baseline treats gestures as strings of discrete symbols and classifies them using an edit-distance. The approach proceeds in two steps.

Vector quantization. Each 3D point $\mathbf{r}(t)$ is first assigned to one of K codewords obtained by clustering the pooled training points (via k-means). A gesture is thereby mapped onto a sequence of discrete symbols $s(t) \in \{1, \dots, K\}$, turning the continuous trajectory problem into a string-matching problem. This discretization step is necessary because edit-distance is fundamentally a string-based metric, designed to compare sequences of discrete elements rather than continuous coordinates.

Edit distance. The dissimilarity between two such symbolic sequences is measured by the Levenshtein edit distance [8], defined as the minimum number of elementary edit operations (insertion, deletion, substitution) required to transform one sequence into the other :

$$d_{\mathrm{ED}}(S_1, S_2) = \min \{\text{cost of operations to transform } S_1 \to S_2\}. \tag{4}$$

The distance is computed using a standard dynamic-programming recursion. Let $D(i, j)$ denote the edit distance between the first i symbols of S_1 and the first j symbols of S_2. Then :

$$D(i,j) = \min \begin{cases} D(i-1,j) + 1 & \text{(deletion)} \\ D(i,j-1) + 1 & \text{(insertion)} \\ D(i-1,j-1) + c(i,j) & \text{(substitution)} \end{cases} \tag{5}$$

where $c(i, j) = 0$ if $S_1[i] = S_2[j]$, and $c(i, j) = 1$ otherwise. The final value $D(|S_1|, |S_2|)$ gives the edit distance between the two complete sequences. Classification is performed with a k-nearest-neighbors rule, assigning each test gesture the label of its k nearest templates in the edit-distance metric.

3.3 The Three-Cents recognizer

The Three-Cents recognizer is a 3D specialization of the \$1 recognizer of Wobbrock, Wilson and Li [13]. The original \$1 algorithm is a widely used gold standard for 2D gesture prototyping, relying on four steps : *resample*, *rotate to a reference angle*, *scale*, and *translate to the origin*, followed by template matching. While effective, its iterative rotation search (Golden-Section Search) becomes computationally expensive and geometrically ambiguous in 3D, where no single reference angle exists.

Caputo et al. [1] propose to forgo the rotation step altogether, yielding a lightweight three-step normalization pipeline (*resample–scale–translate*) that remains robust for short mid-air trajectories.

Three-step normalization pipeline

1. **Resampling.** The raw trajectory is resampled into n equidistant points along its arc length (typically $n = 64$, as in the original \$1 paper). This makes the representation invariant to the execution speed.

2. **Scaling.** The resampled gesture is rescaled using a *uniform* scaling by total arc length. Let L denote the total arc-length of the resampled trajectory :

$$L = \sum_{i=1}^{n-1} \|\mathbf{r}'_{i+1} - \mathbf{r}'_i\|_2. \tag{6}$$

   Following [1], each point is then divided by L :

$$\mathbf{r}''_i = \frac{\mathbf{r}'_i}{L}. \tag{7}$$

   This choice differs from the bounding-box scaling used in the original \$1 algorithm, which stretches each axis independently and can distort the gesture shape. Uniform length-based scaling preserves the proportions of the trajectory, making two gestures of the same shape but different sizes comparable while preserving directional information (crucial for 3D mid-air gestures where orientation is discriminative) [1].

3. **Translation.** Finally, the gesture is translated so that its centroid lies at the origin (0, 0, 0),

$$\mathbf{r}_c = \tfrac{1}{n} \sum_{i=1}^{n} \mathbf{r}'_i, \qquad \mathbf{r}_i^{\text{final}} = \mathbf{r}'_i - \mathbf{r}_c, \tag{8}$$

   ensuring position invariance.

Classification

After normalization, the candidate gesture $C = (\mathbf{c}_1, \dots, \mathbf{c}_n)$ is compared to each template $T = (\mathbf{t}_1, \dots, \mathbf{t}_n)$ by the average Euclidean point-to-point distance :

$$D(C,T) = \frac{1}{n} \sum_{i=1}^{n} \|\mathbf{c}_i - \mathbf{t}_i\|_2. \tag{9}$$

A 1-Nearest-Neighbor rule is then applied : the candidate is assigned the label of the template achieving the minimum distance. By omitting the rotation step and relying on a fixed point-to-point correspondence, the Three-Cents recognizer provides a "cheap" alternative to alignment-based methods such as DTW, at the cost of sensitivity to global rotations.

3.4 Bidirectional LSTM

Unlike the previous methods, which are distance-based, the last approach relies on a recurrent neural network that learns its representation directly from the data. Recurrent Neural Networks (RNNs) process a sequence $\mathcal{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ by maintaining a hidden state

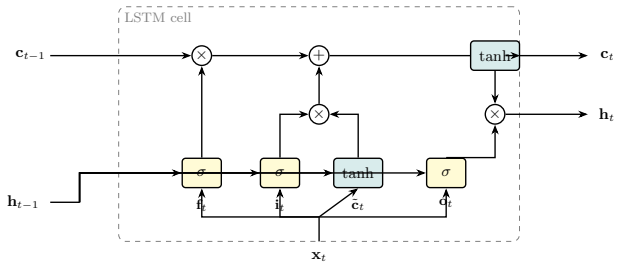

Figure 1 — *Internal data flow of an LSTM cell. Three sigmoid gates (f_t, i_t, o_t) and one* tanh *candidate ($\tilde{c}_t$) jointly update the cell state $\mathbf{c}_t$ and produce the hidden state $\mathbf{h}_t$. The horizontal line carrying $\mathbf{c}_t$ acts as a gradient highway, enabling learning over long sequences.*

that propagates information forward in time. Standard RNNs trained by back-propagation through time suffer however from the *vanishing/exploding gradient problem* : the influence of an early time step on the loss decays (or blows up) exponentially with the temporal distance, making it practically impossible to learn long-range dependencies [4, 3]. Since a gesture trajectory typically spans several dozens of frames whose global shape is the primary discriminating cue, this is a critical limitation.

LSTM cell

The Long Short-Term Memory (LSTM) cell of Hochreiter and Schmidhuber [4] addresses this issue through an internal *cell state* $\mathbf{c}_t$, regulated by three multiplicative gates. At each time step t, given the input $\mathbf{x}_t$ and the previous hidden state $\mathbf{h}_{t-1}$:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f), \tag{10}$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i), \tag{11}$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o), \tag{12}$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c), \tag{13}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \tag{14}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \tag{15}$$

where σ is the logistic sigmoid, $\odot$ the element-wise product, and $\{\mathbf{W}_\bullet, \mathbf{U}_\bullet, \mathbf{b}_\bullet\}$ are the learned parameters. The *forget gate* $\mathbf{f}_t$ controls how much of the previous cell state is preserved, the *input gate* $\mathbf{i}_t$ how much of the candidate $\tilde{\mathbf{c}}_t$ is written to the new state, and the *output gate* $\mathbf{o}_t$ how much of the cell state is exposed as output. The additive update of $\mathbf{c}_t$ creates a *constant error carrousel* through which gradients can flow unattenuated across many time steps, which is the key mechanism that overcomes the vanishing gradient problem [4]. Figure 1 summarizes the cell's internal data flow.

Bidirectional extension

A unidirectional LSTM reads the sequence from left to right and classifies the gesture using only the causal context up to the current frame. Since gesture recognition is an offline task — the full trajectory is available before any decision is made — this is unnecessarily restrictive. The Bidirectional RNN of Schuster and Paliwal [11] runs two independent recurrent layers in opposite directions and concatenates their hidden states :

$$\overrightarrow{\mathbf{h}}_t = \mathrm{LSTM}_{\rightarrow}(\mathbf{x}_1, \dots, \mathbf{x}_t), \quad \overleftarrow{\mathbf{h}}_t = \mathrm{LSTM}_{\leftarrow}(\mathbf{x}_T, \dots, \mathbf{x}_t), \tag{16}$$

$$\mathbf{H}_t = \left[\overrightarrow{\mathbf{h}}_t \,;\, \overleftarrow{\mathbf{h}}_t \right] \in \mathbb{R}^{2h}. \tag{17}$$

Each frame is thus encoded with access to the *entire* temporal context. This is particularly useful when the beginning of a digit is ambiguous before its closing stroke is observed (e.g., a 0 versus a 6, or a 3 versus an 8). In our implementation each direction has h hidden units (hyperparameter `n_units`), and only the final concatenated state $\mathbf{H}_T$ is forwarded to the classifier head (`return_sequences=False`).

Classifier head

The state $\mathbf{H}_T$ is passed through a small feed-forward classifier illustrated in Figure 2. Two regularization mechanisms are applied first. *Batch normalization* [6] rescales each feature of $\mathbf{H}_T$ to zero mean and unit variance over the mini-batch $\mathcal{B}$, before applying a learned affine transformation :

$$\hat{y}_k = \gamma_k \frac{y_k - \mu_{\mathcal{B},k}}{\sqrt{\sigma^2_{\mathcal{B},k} + \varepsilon}} + \beta_k. \tag{18}$$

This reduces the *internal covariate shift* caused by changing upstream weights during training, stabilizes optimization, and allows slightly higher learning rates. *Dropout* [12] is then applied with rate $p = 0.30$: during training only, each activation is independently set to zero with probability p, forcing the network to learn redundant representations and acting as an implicit ensemble of thinned sub-networks. The resulting vector is mapped through a dense layer of 32 ReLU units, then to a softmax over the $K = 10$ classes producing a valid probability distribution $p_k = e^{z_k} / \sum_j e^{z_j}$.

Training

The network is trained by minimizing the categorical cross-entropy

$$\mathcal{L} = -\sum_{k=1}^{K} y_k \log p_k, \tag{19}$$

where $\mathbf{y}$ is the one-hot encoded label. Optimization uses Adam [7], an adaptive first-order method that maintains exponentially decaying estimates of the first ($\mathbf{m}_t$) and second ($\mathbf{v}_t$) moments of the gradient and adjusts the per-parameter step size accordingly. The default hyperparameters ($\alpha = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) are used. As deep models overfit easily on the $\sim$1 000-sample dataset, a 10 % validation split is held out from each training fold to monitor validation loss and *early-stop* training when no improvement is observed for 5 consecutive epochs (best weights restored). Consistent with the other methods, the model is re-initialized from scratch at every cross-validation fold so that no information leaks between folds.

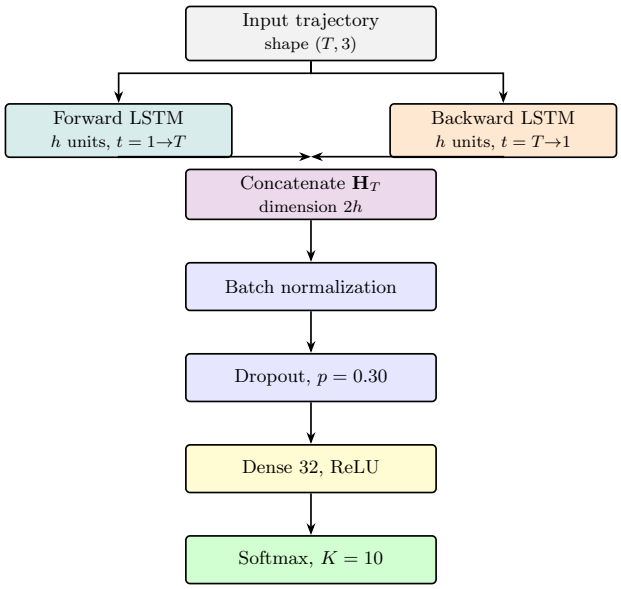

Figure 2 – *Architecture of the BiLSTM classifier. The trajectory (resampled to T points) is processed by two LSTM layers reading in opposite directions ; their final hidden states are concatenated into $\mathbf{H}_T \in \mathbb{R}^{2h}$ and passed through a regularized classifier head producing a softmax over the $K = 10$ gesture classes.*

4 Implementation and methodology

This section describes the data preprocessing, cross-validation protocols, hyperparameter selection, and experimental setup used to evaluate the four gesture recognition methods.

4.1 Data preprocessing

All gestures were loaded from the raw data files and preprocessed in a consistent manner across methods. The preprocessing pipeline consisted of three main steps :

Normalization. Each gesture trajectory was normalized using standardization (z-score normalization) computed exclusively on the training set to prevent data leakage. For each fold, the mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$ were calculated from all training gesture points, and then applied uniformly to both training and test sets :

$$\mathbf{r}_i^{\text{norm}} = \frac{\mathbf{r}_i - \boldsymbol{\mu}}{\boldsymbol{\sigma}}. \tag{20}$$

This removes scale and location artifacts, ensuring that all methods operate on gesture trajectories with zero mean and unit variance.

Dimensionality reduction (optional). To assess the effect of dimensionality reduction, we optionally applied Principal Component Analysis (PCA) to the training set before classification. PCA was fit only on the training data and the resulting transformation was applied to both training and test sets. We tested PCA with $n_{\text{components}} \in \{2, 3\}$ as well as the baseline (no PCA, keeping all 3 dimensions). For gestures with variable-length trajectories, PCA was applied pointwise, preserving the temporal structure.

Variable-length handling. Gesture trajectories in both domains exhibit variable length (number of points). For methods requiring fixed-length input (Three-Cents recognizer), resampling into a fixed number of equidistant points was performed as part of the method-specific preprocessing (described below). DTW and Edit-distance handle variable-length sequences natively without modification.

4.2 Vector quantization for edit-distance

The Edit-distance baseline requires discrete symbolic sequences rather than continuous coordinates. We converted each trajectory into a sequence of cluster labels using k-means clustering, following the project guidelines.

Clustering procedure. For each fold, k-means clustering was fit on all training gesture points (pooled across all gestures and users) with a fixed random seed (42) for reproducibility. We tested cluster counts $K \in \{15, 20, 25, 30, 35, 40, 45, 50, 55, 60\}$ to identify the optimal discretization granularity. After fitting, each point in both training and test sets was assigned to the nearest cluster centroid, transforming each gesture into a string of cluster labels (e.g., $A \to A \to B \to C \to \dots$).

Compression. We also tested an optional compression step that removed consecutive duplicate symbols (e.g., $A \to A \to B$ becomes $A \to B$), reducing edit-distance sensitivity to repetitive gestures. This was tested as a binary option : compression enabled or disabled.

4.3 Cross-validation and hyperparameter selection

All methods were evaluated using two complementary cross-validation protocols to assess both user-independence and generalization within users.

User-independent evaluation. We performed leave-one-user-out cross-validation : for each of the 10 subjects in turn, we held out all gestures from that subject as the test set and trained on all gestures from the remaining 9 subjects. This was repeated 10 times (once per subject), yielding 10 accuracy measurements per method/domain/hyperparameter combination.

User-dependent evaluation. We performed leave-one-gesture-sample-out cross-validation : for each user and each gesture type, we iteratively held out one sample (repetition) as the test set, reserving the other 9

repetitions from that user for training. This was repeated 10 times (once per repetition index), again yielding 10 accuracy measurements per configuration.

Inner validation split for hyperparameter selection. To avoid any information leakage from the outer test fold into the hyperparameter search, we introduced a nested validation step. Within each outer fold, the training portion is further split randomly into an *inner training* set (80 %) and an *inner validation* set (20 %), using a fixed random seed (42) for reproducibility :

$$\mathcal{D}_{\text{train}}^{\text{outer}} \xrightarrow{80/20} \mathcal{D}_{\text{inner-train}} \cup \mathcal{D}_{\text{inner-val}}. \tag{21}$$

All preprocessing statistics (mean, standard deviation, k-means centroids, PCA components) and the method parameters are fitted *exclusively* on $\mathcal{D}_{\text{inner-train}}$. Candidate hyperparameter configurations are then scored on $\mathcal{D}_{\text{inner-val}}$. The outer test fold is thus *never* seen during hyperparameter selection, ensuring an unbiased performance estimate. Once the best configuration is identified, the full outer training set $\mathcal{D}_{\text{train}}^{\text{outer}}$ is used for the final model fit before evaluating on the held-out test fold.

Hyperparameter grid search. Within each outer fold, we performed a grid search over the inner validation set, covering hyperparameters specific to each method :

- **Edit-distance :** $K \in \{15, 20, 25, 30, 35, 40\}$ (cluster count), compression $\in$ {True, False}, and $k \in \{1, 3, 5, 7\}$ for k-NN.
- **DTW :** $k \in \{1, 3, 5, 7\}$ for k-NN ; no method-specific hyperparameters.
- **Three-Cents :** resampling to $n_{\text{points}} \in \{16, 32, 64\}$ equidistant points.
- **PCA :** $n_{\text{components}} \in \{\text{no PCA}, 2, 3\}$ applied to all methods.

For each outer fold, we computed inner-validation accuracy for all hyperparameter combinations. The best configuration per (method, domain, cv_mode) tuple was selected by maximizing mean inner-validation accuracy across folds, and then applied to the corresponding outer test fold.

4.4 Classification and distance metrics

k-Nearest neighbors (k-NN). DTW and Edit-distance both use k-NN for classification. For each test gesture, we computed distances to all training gestures, sorted the results, and assigned the test gesture the majority-vote label among its k nearest neighbors. We tested $k \in \{1, 3, 5, 7\}$ to balance between local (small k) and global (large k) neighborhood effects.

Distance computation. For DTW, we used the accumulated dynamic programming distance $D(N, M)$ as the distance metric. For Edit-distance, we used the Levenshtein edit distance computed via dynamic programming. For Three-Cents, we used the average point-to-point Euclidean distance.

Three-Cents 1-NN. The Three-Cents recognizer uses 1-nearest-neighbor by design ; the k parameter was fixed at 1 for this method.

4.5 Implementation details and reproducibility

All methods were implemented in Python 3 using NumPy for numerical computation. DTW and Edit-distance distance computations were implemented from scratch (without external gesture recognition libraries) as required by the project guidelines. PCA and k-means clustering were sourced from scikit-learn.

Random seeds. The k-means clustering algorithm was initialized with random seed 42, and 10 different centroid initializations were tested per fold (n_init=10) to ensure robust cluster center identification.

Data leakage prevention. Critical to the experimental integrity : normalization statistics (mean, std) and clustering centroids were learned *only* from training data and applied to test data, preventing information leakage from test to train.

Metrics. Classification accuracy was computed as the proportion of correct predictions :

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of test samples}}. \tag{22}$$

For each configuration, we report mean accuracy and standard deviation across the 10 cross-validation folds.

5 Results and discussion

This section presents and discusses the experimental results obtained by the four classifiers on domain 1 (digits) and domain 4 (3D figures), under both the user-independent and the user-dependent cross-validation protocols. We first report the best hyperparameter configurations selected on the inner-validation folds, then the test accuracies, the statistical comparison of the methods, and finally a per-class confusion analysis of the best model.

5.1 Best hyperparameter configurations

For every (method, domain, CV-mode) combination, the configuration achieving the highest mean inner-validation accuracy was retained and re-fitted on the full outer training fold. The resulting selections are reported in Table 1. They are remarkably stable across

the four experimental settings, which suggests that the chosen grids cover sensible operating points and that the selected configurations do not overfit the inner-validation set.

Table 1 – *Best hyperparameter configurations selected by inner-validation, per method, domain and CV protocol. "–" means the hyperparameter does not apply.*

Method	Hyperparameter	Domain 1 (digits)		Domain 4 (3D figures)	
		user-indep.	user-dep.	user-indep.	user-dep.
DTW	PCA	none	none	none	none
	k-NN neighbours k	1	1	1	1
Edit-distance	PCA	none	none	none	none
	Codebook size K	40	35	20	40
	Compression	No	No	No	No
	k-NN neighbours k	1	1	1	1
Three-Cents	PCA	**2D**	**2D**	none	none
	Resampling points n	16	16	16	16
BiLSTM	Sequence length T	64	16	16	64
	Hidden units h	64	64	32	64

Several observations can be drawn from Table 1. For the two dynamic-programming baselines, the optimal k-NN neighbourhood size is always $k = 1$: the closest single template is more discriminative than a larger neighbourhood, indicating that the inter-class distances are large relative to intra-class variability once a proper sequence metric is used. Reducing the dimensionality of the trajectories with PCA never improves the DTW or Edit-distance accuracies, which is expected since both methods already exploit the full 3D geometry of the signal. For the Edit-distance baseline, the selected codebook sizes ($K \in \{20, 35, 40\}$) cluster around the middle of the grid ; smaller values would produce overly coarse strings, while larger values fragment the same trajectory shape into too many symbols and inflate the edit cost. Disabling the compression of consecutive duplicate symbols was systematically preferred : compression discards local timing information, which apparently still carries discriminative signal even after vector quantization.

The Three-Cents recognizer consistently selects $n = 16$ resampling points, which is markedly lower than the $n = 64$ value used in the original \$1 recognizer [13]. This is consistent with the short and relatively simple trajectories of the considered domains : too dense a resampling adds noise without bringing new geometric information. Interestingly, the recognizer benefits from a 2D PCA projection on domain 1 (planar digits) but prefers the full 3D coordinates on domain 4 (geometric figures, which are intrinsically three-dimensional). This is precisely the behaviour one would expect from a method that relies on a fixed point-to-point correspondence : when the signal is intrinsically planar, removing the noisy out-of-plane component sharpens the comparison.

Finally, the BiLSTM selects the smallest hidden state ($h = 32$) in the hardest setting (domain 4, user-independent) and the largest ($h = 64$) elsewhere, which is consistent with the well-known trade-off between model capacity and overfitting on a $\sim$1 000-sample dataset. The selected sequence length T varies between 16 and 64 frames without a clear pattern, suggesting that the model is relatively robust to this choice.

5.2 Ablation Study

This section isolates the effect of each preprocessing decision (z-score normalization and dimensionality reduction via PCA ($n_{\text{components}} \in \{2, 3\}$ or none)) on the test accuracy of the four classifiers, across both domains and both cross-validation protocols. The study covers all 80 configurations systematically.

5.2.1 Effect of Normalization

General trend. Normalization brings a systematic but modest gain in the user-dependent regime (+0.2 to +3.0 pp depending on method and domain) and a more substantial gain in the user-independent regime (+1.5 to +20.8 pp for DTW, +1.6 to +17.6 pp for edit-distance). This contrast has a straightforward mechanical explanation : in user-dependent mode, training templates come from the same user as the test sample, so absolute amplitudes and positions are already coherent, and normalization adds little. In user-independent mode, raw trajectories can differ substantially across individuals in amplitude and centering ; standardization reduces this covariate shift and improves generalization.

DTW. The effect is particularly pronounced on domain 4 in independent mode : normalization alone (without PCA) moves accuracy from 55.2% to 60.0% (+4.9 pp), and combined with PCA-2D, from 57.4% to 78.0% (+20.6 pp). This last figure is the most dramatic gain in the entire study. However, the standard deviation remains high ($\sigma \approx 0.08$–0.09), which tempers enthusiasm.

Edit-distance. The gain from normalization is systematic but non-uniform across PCA settings. On domain 1 independent, normalization improves accuracy by +2.0 pp (without PCA), +4.8 pp (PCA-2D), and +4.1 pp (PCA-3D). On domain 4 the gains are even more variable : +8.9 pp (without PCA), +17.5 pp (PCA-2D), +12.8 pp (PCA-3D). This interaction between normalization and PCA suggests that the two transformations are not independent : normalization conditions the geometry of the space in which PCA operates, and the effectiveness of the subsequent vector quantization depends directly on this.

Three-Cents. The effect of normalization is remarkably weak for this classifier. On domain 1 independent, the difference between unnormalized and normalized is 0.0 pp (without PCA), +0.2 pp (PCA-2D), and +0.5 pp (PCA-3D) all below the inter-fold standard deviation and therefore statistically indistinguishable from zero. This result is not surprising : the Three-Cents method already incorporates intrinsic geometric normalization which absorbs most of the amplitude and position variability that z-score standardiza-

tion is meant to correct. In other words, normalization is largely redundant with the internal invariances of Three-Cents. This is a non-trivial ablation finding, and it suggests that the design choices of the Three-Cents recognizer are well-motivated from a preprocessing perspective.

BiLSTM. Normalization has the largest relative effect here : on domain 1 dependent, it moves accuracy from 95.2% to 98.4% (+3.2 pp), and on domain 4 dependent, from 85.8% to 90.9% (+5.1 pp). In independent mode the effect is less clear-cut (+2.4 pp on domain 1, −1.2 pp on domain 4, slightly negative but within the margin of error). This heightened sensitivity of the BiLSTM to normalization is consistent with the broader literature on recurrent networks : LSTMs are known to be sensitive to input scale, and the absence of normalization can induce gradient instabilities or convergence to suboptimal minima. The absence of PCA configurations for the BiLSTM in our experiments constitutes a limitation : we cannot assess whether dimensionality reduction would provide additional regularization for this classifier.

5.2.2 Effect of Dimensionality Reduction (PCA)

DTW. PCA systematically improves DTW in independent mode, but non-monotonically with respect to the number of components. On domain 1, the best test accuracy is obtained with PCA-2D normalized (83.6%) rather than PCA-3D normalized (82.6%), whereas one might intuitively expect more dimensions to preserve more discriminant information. This counterintuitive result can be explained by DTW's sensitivity to noise in each dimension independently. By projecting onto two components, the third component (likely dominated by depth (z) noise that is poorly reproducible across users) is discarded. PCA thus acts not as an information reduction but as implicit denoising.

On domain 4, the pattern is cleaner : PCA-2D normalized (78.0%) outperforms PCA-3D normalized (74.7%), both well above no-PCA (60.0%). Here the depth component of the 3D figures may also carry inter-user noise that the 2D projection filters out, despite these figures being intrinsically 3D. This is a somewhat surprising finding and suggests that for DTW on domain 4, a 2D PCA is not simply discarding irrelevant information but is genuinely improving the signal-to-noise ratio.

Edit-distance. PCA improves edit-distance significantly only when combined with normalization. On domain 4 independent, PCA-2D without normalization gives 45.7%, versus 63.2% with normalization, a difference of 17.5 pp that does not come from PCA alone but from the interaction of both transformations. More striking, PCA-3D without normalization (47.2%) is barely above no-PCA without normalization (44.2%) : PCA adds virtually nothing, or even marginally degrades, without prior normalization. This highlights a genuine methodological risk : applying PCA to non-normalized data amounts to projecting onto directions dominated by high-variance dimensions (typically z, the depth axis), which are not necessarily the most discriminant ones. The practical implication is clear : PCA should never be applied to raw gesture data without prior normalization.

Three-Cents. The contrast between domain 1 and domain 4 is particularly instructive. On domain 1, PCA brings little in dependent mode (no-PCA configurations already reach 100%) but a modest gain in independent mode : PCA-3D normalized reaches 97.7% vs. 95.8% without PCA (+1.9 pp). On domain 4, the gain is much larger : PCA-2D normalized reaches 93.8% in independent mode vs. 84.8% without PCA (+9.0 pp), while PCA-3D normalized (95.1%) is the overall best configuration.

This contrast reflects the intrinsic geometry of the two domains. Digits (domain 1) are essentially planar figures drawn in the air : their inter-user variability concentrates on the third dimension (z), which is geometrically non-informative. PCA-2D eliminates precisely this noisy dimension, consistent with the selection of PCA-2D by inner validation in our main experiments. The geometric figures (domain 4) are intrinsically 3D ; PCA-3D retains all discriminant directions while removing residual noise through standardization. The fact that PCA-3D (which involves no actual dimensionality reduction on 3D data) improves results suggests that it is the normalization rather than the projection itself that carries most of the gain, with PCA here playing a role of axis decorrelation rather than dimension reduction.

5.2.3 Summary

Three main conclusions emerge from this ablation study.

1. **Normalization is quasi-universally beneficial**, but its magnitude varies by an order of magnitude across methods. It is essential for DTW, edit-distance, and BiLSTM in independent mode, but redundant for Three-Cents due to its internal invariances.
2. **PCA is a double-edged tool**. Combined with normalization, it significantly improves DTW and edit-distance. Without normalization, it can degrade performance, particularly for edit-distance on domain 4. Its effect is never monotone with respect to the number of components, which argues for careful and context-specific use rather than systematic application.
3. **Three-Cents is remarkably robust to preprocessing choices**. The gap between its worst and best configuration in independent mode is 1.9 pp (domain 1) and 13.8 pp (domain 4). This robustness is a considerable practical advantage, though it also means that preprocessing optimi-

zation offers limited leverage for this classifier beyond the baseline.

5.3 Classification accuracy

Table 2 reports the mean test accuracy and standard deviation across the 10 cross-validation folds for each method, domain and CV protocol, using the best configuration identified in Section 5.1.

Table 2 – *Mean test accuracy (%) ± standard deviation across 10 folds. Best result per column in* ***bold****.*

	Domain 1		Domain 4	
Method	indep.	dep.	indep.	dep.
DTW	82.6 ± 12.4	99.6 ± 0.5	74.4 ± 11.2	99.5 ± 1.0
Edit-distance	67.9 ± 23.1	99.1 ± 0.7	60.2 ± 19.5	98.3 ± 1.4
Three-Cents	$\mathbf{96.3 \pm 4.6}$	$\mathbf{99.8 \pm 0.4}$	$\mathbf{95.1 \pm 5.3}$	$\mathbf{98.6 \pm 1.3}$
BiLSTM	85.0 ± 12.7	97.2 ± 2.2	70.1 ± 8.2	88.5 ± 3.2

Three main patterns emerge from Table 2.

User-dependent accuracy is systematically higher than user-independent accuracy for every method and every domain, confirming the expected answer to (Q2). The gain is particularly dramatic for the two dynamic-programming baselines : DTW improves by +17.0 and +25.1 percentage points on domains 1 and 4 respectively, while Edit-distance improves by +31.2 and +38.1 points. When the model has access to several repetitions of the same gesture by the same user, the 1-NN rule becomes a near-trivial matching task, which explains the near-perfect user-dependent scores ($\geq 98\%$) of DTW, Edit-distance and Three-Cents. The residual gap observed for BiLSTM (+12.2 and +18.4 points) is comparatively small, confirming that the main limitation of the neural approach is not the user-shift per se, but the insufficient training-set size.

Three-Cents is the best method in the user-independent setting on both domains, achieving 96.3% on domain 1 and 95.1% on domain 4 with a low standard deviation ($\leq 5.3\%$), pointing to (Q1). This result is remarkable given the simplicity of the method : by omitting the rotation step of the original \$1 recognizer and relying on a lightweight resample–scale–translate pipeline, Three-Cents outperforms all competitors by a wide margin. The runner-up is BiLSTM on domain 1 (85.0%) and DTW on domain 4 (74.4%), both well below Three-Cents.

Edit-distance is the weakest method in the user-independent setting and exhibits the largest variability (std up to 23.1%). The vector quantization step, necessary to convert continuous 3D trajectories into discrete symbol strings, entails a lossy compression of the geometric information. When templates come from different users (user-independent case), this loss is compounded by inter-user style variability, leading to inconsistent codebook assignments and poor generalization. In the user-dependent setting, however, Edit-distance nearly saturates ($\geq 98\%$), showing that the method is not fundamentally flawed but simply sensitive to the covariate shift introduced by unseen users.

5.4 Confusion Matrix Analysis

The Three-Cents method represents the best-performing method across all settings. The confusions that emerge are geometrically motivated in every case : misclassifications arise when two gesture classes share enough trajectory structure to be ambiguous under the resample–scale–translate normalization, particularly when the model must generalize across users.

5.4.1 Domain 1 : Digits (0–9)

User-dependent (99.8% accuracy). Near-perfect recognition. The only confusion is 2 samples of digit 0 misclassified as digit 8. This is the single most intuitive mistake in the entire dataset : both gestures share a closed oval shape, and their 3D trajectories differ mainly in whether a second loop is drawn. In a user-dependent setting where the model has seen each user's style repeatedly, even this pair is almost perfectly separated. Every other digit achieves 100/100.

Table 3 – *Confusion matrix – Domain 1, user-dependent (99.8%).*

	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
0	98	0	0	0	0	0	0	0	2	0
1	0	100	0	0	0	0	0	0	0	0
2	0	0	100	0	0	0	0	0	0	0
3	0	0	0	100	0	0	0	0	0	0
4	0	0	0	0	100	0	0	0	0	0
5	0	0	0	0	0	100	0	0	0	0
6	0	0	0	0	0	0	100	0	0	0
7	0	0	0	0	0	0	0	100	0	0
8	0	0	0	0	0	0	0	0	100	0
9	0	0	0	0	0	0	0	0	0	100

User-independent (96.3% accuracy). The model generalizes well across users, but two pairs reveal inter-user style variability. Digit 0 is the most problematic class : 10 out of 100 samples are misclassified as digit 3, 6 as digit 6, and 8 as digit 8. This is geometrically coherent : digit 0 shares a circular arc with 6 and 8, and some users draw 0 with a slight leftward lean that resembles 3. Conversely, digit 8 also sees 7 of its own samples confused back with 0, confirming that the $0 \leftrightarrow 8$ boundary is the hardest in the domain. A minor $9 \rightarrow 7$ confusion (1 sample) and a $5 \rightarrow 2$ confusion (1 sample) are present but negligible.

Table 4 – *Confusion matrix – Domain 1, user-independent (96.3%).*

	0	1	2	3	4	5	6	7	8	9
0	72	0	0	10	4	0	6	0	8	0
1	0	100	0	0	0	0	0	0	0	0
2	0	0	100	0	0	0	0	0	0	0
3	0	0	0	100	0	0	0	0	0	0
4	0	0	0	0	100	0	0	0	0	0
5	0	0	1	0	0	99	0	0	0	0
6	0	0	0	0	0	0	100	0	0	0
7	0	0	0	0	0	0	0	100	0	0
8	7	0	0	0	0	0	0	0	93	0
9	0	0	0	0	0	0	0	1	0	99

The asymmetry is noteworthy : digit 0 is misclassified into several other classes (3, 4, 6, 8), while those classes rarely mistake themselves for a 0. This suggests that 0 is the most geometrically ambiguous digit : its circular shape has the highest overlap with the trajectory space of other digits when drawn by unseen users.

5.4.2 Domain 4 : 3D Geometric Figures

User-dependent (98.6% accuracy). A few localized confusions appear. The main one is Pyramid being confused with Cylindrical Pipe (6/100), which reflects that both involve elongated swept-stroke trajectories without strong closed-shape anchors. Cylinder loses 2 samples to Rectangular Pipe and 1 to Toroid, which makes sense as all three involve a dominant circular cross-section. Hemisphere loses 2 samples to Cuboid and 1 to Sphere, consistent with the ambiguity between a half-sphere and a full sphere or a boxy gesture.

Table 5 – *Confusion matrix – Domain 4, user-dependent (98.6%).*

	Cu.	Cy.	Sph.	R. Pipe	Hemi.	Cone	C. Pipe	Py.	Te.	To.
Cuboid	100	0	0	0	0	0	0	0	0	0
Cylinder	0	100	0	0	0	0	0	0	0	0
Sphere	0	0	98	1	0	0	0	0	0	1
Rect. Pipe	0	0	0	100	0	0	0	0	0	0
Hemisphere	2	0	1	0	97	0	0	0	0	0
Cone	0	0	0	0	0	100	0	0	0	0
Cyl. Pipe	0	0	0	0	0	0	98	2	0	0
Pyramid	0	0	0	1	0	0	6	93	0	0
Tetrahedron	0	0	0	0	0	0	0	0	100	0
Toroid	0	0	0	0	0	0	0	0	0	100

User-independent (95.1% accuracy). Confusions intensify and spread. The most dramatic is Cone → Rectangular Pipe : 12 out of 100 Cone samples are misclassified, reflecting that the tapering apex of a cone can resemble the elongated rectangular sweep of a pipe when drawn mid-air with inter-user variability in stroke order and depth. Pyramid → Cylindrical Pipe worsens to 16 misclassifications (the strongest off-diagonal entry in the entire dataset), suggesting these two are the hardest pair in Domain 4. Cylinder and Hemisphere remain slightly confused with neighboring classes.

Table 6 – *Confusion matrix – Domain 4, user-independent (95.1%).*

	Cu.	Cy.	Sph.	R. Pipe	Hemi.	Cone	C. Pipe	Py.	Te.	To.
Cuboid	100	0	0	0	0	0	0	0	0	0
Cylinder	0	98	0	0	0	0	0	0	2	0
Sphere	0	0	97	1	0	1	0	0	0	1
Rect. Pipe	0	0	0	100	0	0	0	0	0	0
Hemisphere	2	0	1	0	97	0	0	0	0	0
Cone	0	0	0	12	0	87	0	0	0	1
Cyl. Pipe	0	0	0	0	0	0	93	7	0	0
Pyramid	0	0	0	0	0	1	16	83	0	0
Tetrahedron	0	0	0	1	0	0	0	0	97	2
Toroid	0	0	1	0	0	0	0	0	0	99

The Pyramid–Cylindrical Pipe confusion warrants further discussion. Both gestures involve a dominant vertical or diagonal stroke followed by shorter strokes, and neither has a strongly closed circular or rectangular element that would anchor the Three-Cents template. After normalization, the point clouds of these two classes can become nearly indistinguishable when drawn by an unseen user with a different aspect ratio or stroke angle.

5.4.3 Confusion Matrices Summary

Across all four matrices, three patterns hold consistently.

First, **all errors are geometrically motivated**. Every misclassification has a clear geometric explanation. Classes that share a dominant shape element (closed loop, elongated sweep, absent closed base) are always the ones confused. Random or arbitrary errors are entirely absent from the matrices.

Second, **user-shift amplifies but does not create confusion pairs**. The user-independent matrices show the same pairs confused as the user-dependent ones, but at higher rates and with additional secondary confusions. The underlying ambiguity is already present in the user-dependent setting ; cross-user variability simply amplifies it. This suggests that targeted data augmentation simulating inter-user style variation could specifically address these boundaries.

Third, **Domain 4 is structurally harder than Domain 1**. Domain 4 exhibits more widespread confusions in the user-independent setting (8 non-zero off-diagonal entries vs. 5 for Domain 1). The intrinsically 3D nature of geometric figures makes inter-user normalization harder. A digit is drawn the same way by virtually all users ; a Cone or Pyramid admits many valid stroke sequences that produce very different normalized trajectories.

5.5 Statistical comparison of the methods

Statistical tests were conducted on the user-independent setting, which is the most practically relevant scenario and the one addressed by research question (Q3). Following the recommendation of Demšar [2] for comparing multiple classifiers across several datasets, we first applied the non-parametric **Friedman**

test to assess whether at least one method differs significantly from the others (null hypothesis : all methods share the same expected rank). The test was highly significant in both domains ($\chi^2 = 23.03$, $p < 0.001$ for domain 1 ; $\chi^2 = 24.15$, $p < 0.001$ for domain 4), firmly rejecting the null hypothesis and justifying post-hoc pairwise comparisons.

Post-hoc analysis was carried out with two complementary procedures. The **Nemenyi test** computes a critical difference CD = 1.48 (at $\alpha = 0.05$, $k = 4$ methods, $N = 10$ folds) : two methods are declared significantly different if and only if their average Friedman ranks differ by more than CD. Results are visualised in Figure 3, where methods connected by a horizontal bar are *not* significantly different. The **Wilcoxon signed-rank test** with Holm–Bonferroni correction was additionally applied to each pair, providing fold-level evidence that is more sensitive than the rank-based Nemenyi test.

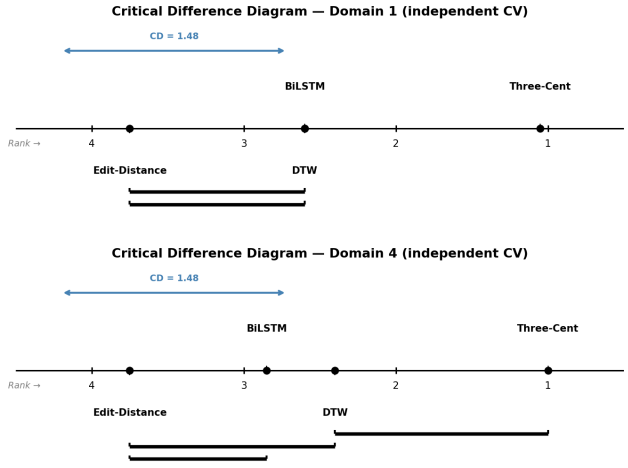

Figure 3 – *Critical difference diagrams (Nemenyi, $\alpha = 0.05$) for domain 1 (top) and domain 4 (bottom) under user-independent CV. Methods are ranked from best (right, rank 1) to worst (left, rank 4). Horizontal bars connect methods whose rank difference does not exceed* CD = 1.48*, i.e. whose performance is* not *statistically distinguishable.*

The diagrams lead to the following conclusions regarding (Q3).

On **domain 1**, Three-Cents (avg. rank 1.05) stands isolated on the right, connected to no other method : it is significantly better than DTW (Wilcoxon adj-$p = 0.016$), Edit-distance (adj-$p = 0.012$) and BiLSTM (adj-$p = 0.010$). DTW and BiLSTM share the same average rank (2.60) and are not distinguishable from each other (adj-$p = 0.625$). The Wilcoxon–Holm procedure does however separate Edit-distance from both DTW (adj-$p = 0.031$) and BiLSTM (adj-$p = 0.023$), a difference the less powerful Nemenyi test cannot detect.

On **domain 4**, Three-Cents (avg. rank 1.00) is again the sole top-ranked method, significantly outperforming Edit-distance (adj-$p = 0.010$), DTW (adj-$p = 0.012$) and BiLSTM (adj-$p = 0.008$). The three remaining methods form a partially indistinct group : Edit-distance and BiLSTM are not separable under either test, and DTW versus BiLSTM is also non-significant (adj-$p = 0.238$). DTW does however significantly outperform Edit-distance under Wilcoxon–Holm (adj-$p = 0.012$).

In summary, **Three-Cents is the only method that is statistically superior to all others in both domains** under the user-independent protocol, providing a clear and unambiguous answer to (Q3). No other pairwise ordering is consistently significant across both domains.

6 Conclusion

Références

[1] Fabio M. Caputo, Pietro Prebianca, Alessandro Carcangiu, Lucio D. Spano, and Andrea Giachetti. Comparing 3d trajectories for simple mid-air gesture recognition. *Computers & Graphics*, 73 :17–25, 2018.

[2] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7 :1–30, 2006.

[3] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016. Available at http://www.deeplearningbook.org.

[4] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8) :1735–1780, 1997.

[5] Jie Huang, Shubham Jaiswal, and Rahul Rai. Gesture-based system for next generation natural and intuitive interfaces. *Artificial Intelligence for Engineering Design, Analysis and Manufacturing*, 33(1) :54–68, 2019.

[6] Sergey Ioffe and Christian Szegedy. Batch normalization : Accelerating deep network training by reducing internal covariate shift. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 448–456, 2015.

[7] Diederik P. Kingma and Jimmy Ba. Adam : A method for stochastic optimization. In *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*, 2015.

[8] Vladimir I. Levenshtein. Binary codes capable of correcting deletions, insertions, and reversals. *Soviet Physics—Doklady*, 10(8) :707–710, 1966.

[9] Cory S. Myers and Lawrence R. Rabiner. A comparative study of several dynamic time-warping algorithms for connected-word recognition. *The Bell System Technical Journal*, 60(7) :1389–1409, 1981.

[10] Lawrence Rabiner and Biing-Hwang Juang. *Fundamentals of Speech Recognition*. Prentice-Hall, Englewood Cliffs, NJ, 1993.

[11] Mike Schuster and Kuldip K. Paliwal. Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 45(11) :2673–2681, 1997.

[12] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout : A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1) :1929–1958, 2014.

[13] Jacob O. Wobbrock, Andrew D. Wilson, and Yang Li. Gestures without libraries, toolkits or training : a $1 recognizer for user interface prototypes. In *Proceedings of the 20th Annual ACM Symposium on User Interface Software and Technology (UIST)*, pages 159–168. ACM, 2007.

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